

lecture. 6

Ex 20
P39 Find Negation.

there is an honest politician.

there exist x , x is a politician.
 x is honest.

$x \in \text{Set of Politician}$
 $p(x) \equiv x \text{ is a honest}$

$$\neg \exists x p(x) \equiv \forall x \neg p(x)$$

for all x , x is a politician.
 x is not honest.

Ex 21 :- $\forall x (x^2 > x)$ Find Negation.
P39

$$\text{let } p(x) \equiv x^2 > x.$$

$$\neg \forall x p(x) \equiv \exists x \neg p(x).$$

$$\neg (>) \equiv \leq$$

$$\equiv \exists x \neg (x^2 > x)$$

$$\equiv \exists x x^2 \leq x.$$

(Ex 1-40
P 43-46)

Ex 22
P44 Every Speaks hindi.

for all x , x is an Indian.
 x Speaks hindi

$x \in \text{Set of Indian.}$

Ex 35
P45 $\forall x (x^2 > 2x)$. $x \in \mathbb{Z}$.
let $p(x) \equiv x^2 > 2x$.
No counter Example.

Nested Quantifiers.

$$\forall x \forall y p(x, y) \equiv \forall x [p(x, 1) \wedge p(x, 2) \wedge \dots \wedge p(x, N)]. \quad x, y \in \{1, 2, \dots, N\}$$

$$\begin{aligned} \forall x \forall y \, p(x, y) &= \forall x \left[p(x, 1) \wedge p(x, 2) \wedge \dots \wedge p(x, N) \right]. \quad x, y \in \{1, 2, \dots, N\} \\ &= \forall x \underline{p(x, 1)} \wedge \forall x \underline{p(x, 2)} \wedge \dots \wedge \forall x \underline{p(x, N)}. \\ &= p(1, 1) \wedge p(2, 1) \wedge p(3, 1) \wedge \dots \wedge p(N, 1) \wedge \\ &\quad p(1, 2) \wedge p(2, 2) \wedge p(3, 2) \wedge \dots \wedge p(N, 2) \wedge \\ &\quad \vdots \\ &\quad p(1, N) \wedge p(2, N) \wedge \dots \wedge p(N, N). \end{aligned}$$

$$\begin{aligned} & \forall x \exists y p(x, y) \\ & \quad \underline{\forall x p(x, 1)} \vee \underline{\forall x p(x, 2)} \vee \dots \vee \underline{\forall x p(x, N)}. \\ & = (p(1, 1) \wedge p(2, 1) \wedge p(3, 1) \wedge \dots \wedge p(N, 1)) \vee \\ & \quad (p(1, 2) \wedge p(2, 2) \wedge p(3, 2) \wedge \dots \wedge p(N, 2)) \vee \\ & \quad \vdots \\ & \quad (p(1, N) \wedge p(2, N) \wedge \dots \wedge p(N, N)). \end{aligned}$$

$$\begin{array}{l} \exists x \forall y \quad P(x,y) \quad ? \\ \exists x \exists y \quad P(x,y) \quad ? \end{array} \quad \text{HW.}$$

$$\begin{array}{ll} \neg \forall x \forall y p(x, y) & \exists x \exists y \neg p(x, y) = ? \\ \neg \forall x \exists y p(x, y) & ? = ? \\ \neg \exists x \forall y p(x, y) & ? = ? \\ \neg \exists x \exists y p(x, y) & \forall x \forall y \neg p(x, y) = ? \end{array}$$

Ex 1 :- HxHy (x+y = y+x). $x, y \in \mathbb{R}$.

Let $P(x, y) : x+y = y+x$.

HxHy $P(x, y) : ?$

$$\begin{aligned} \exists x \forall y \quad Q(x, y) &= x + y \geq 0. \quad x, y \in \mathbb{R}. \\ \exists y \forall x \quad Q(x, y) &= \mathbb{R} \\ \forall x \exists y \quad Q(x, y) &= \mathbb{R} \end{aligned}$$

$$\exists y \forall x Q(x, y) = \text{True}$$

$$\forall x \exists y Q(x, y) = \text{True}$$

the order of Quantifiers does matter.

Ex 5 :-
p 49

$$Q(x, y, z) = x + y = z$$

$$x, y, z \in \mathbb{R}$$

$$\forall x \forall y \exists z Q(x, y, z) = \text{True}$$

$$\exists z \forall x \forall y Q(x, y, z) = \text{False}$$

Ex 7 :-

$$\text{So. } \forall x (x \neq 0) \rightarrow \exists y (xy = 1) \quad x, y \in \mathbb{R}$$

$$\forall x \exists y (x \neq 0) \rightarrow (xy = 1)$$

$$\text{Ex 9 :- } \forall x (C(x) \vee \exists y (C(y) \wedge F(x, y)))$$

$C(a)$: a has a Computer System
 $F(a, b)$: a and b are friends

$$\forall x \exists y (C(x) \vee (C(y) \wedge F(x, y)))$$

for all x, there exist y,
x & y are persons.

$$x, y \in \text{Persons}$$

x has a Computer System or -
y has a " " and.

x and y are friends.

Ex 11:- If a person is a Female and is a Parent

then this person is Saneous mother.

for all x, x is a person.
if x is a female and

there exist y , y is a person,
 x is the parent of y ,

then x is the mother of y .

$x, y \in \text{Set of persons.}$

$f(x) \hat{=} x$ is a female.
 $p(x, y) \hat{=} x$ is a parent of y .

$M(x, y) \hat{=} x$ is the mother of y .

for all x , x is a person.

there exist y , y is a person,

if x is a female and
 x is the parent of y ,

then x is the mother of y .

$$\forall x \exists y (f(x) \wedge p(x, y) \rightarrow M(x, y)).$$

Everybody loves somebody.

for all x , there exist y ,
 x & y are persons,
 x loves y .

$x, y \in \text{Persons.}$
 $L(x, y) \hat{=} x$ loves y .

$$\forall x \exists y L(x, y).$$

Ex 1-40 (PS4-57), HW.

Ex 17
 PS6.

Every student has access to exactly one mail box.

for all x , there exist y , there exist z
 x is a student, y, z is a mail box.

if x has y , then x has z .
 and (y and z are different)

$H(x, y) \hat{=} x$ has y .

1) System Consistency.

2) Duplicate / Converse / Contra / Javers.

- 2) ^U Implication / ^U Converse / Contra / Invers.
- 3) Quantifiers.
- 4) ~~1~~ Know / Knights (Logic puzzles).